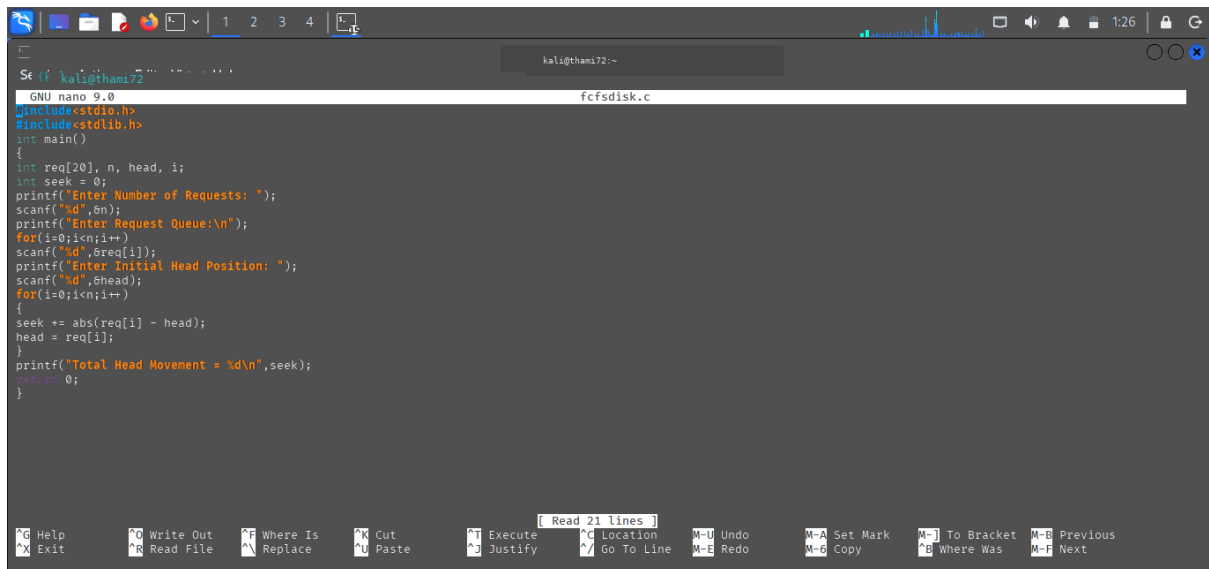
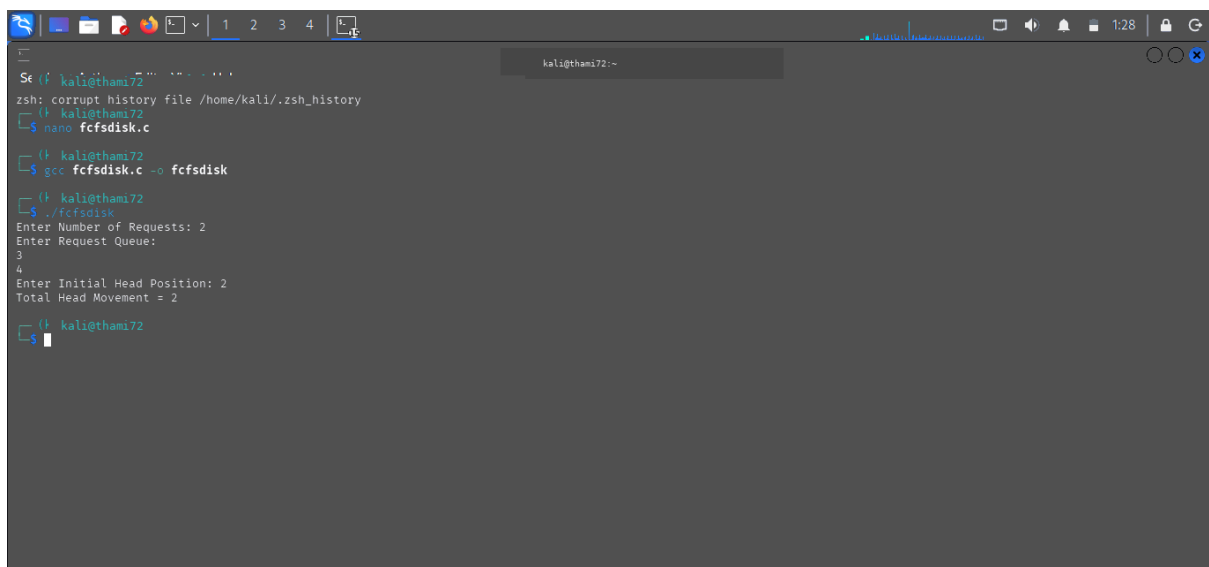


C PROGRAMMING :FCFS DISK



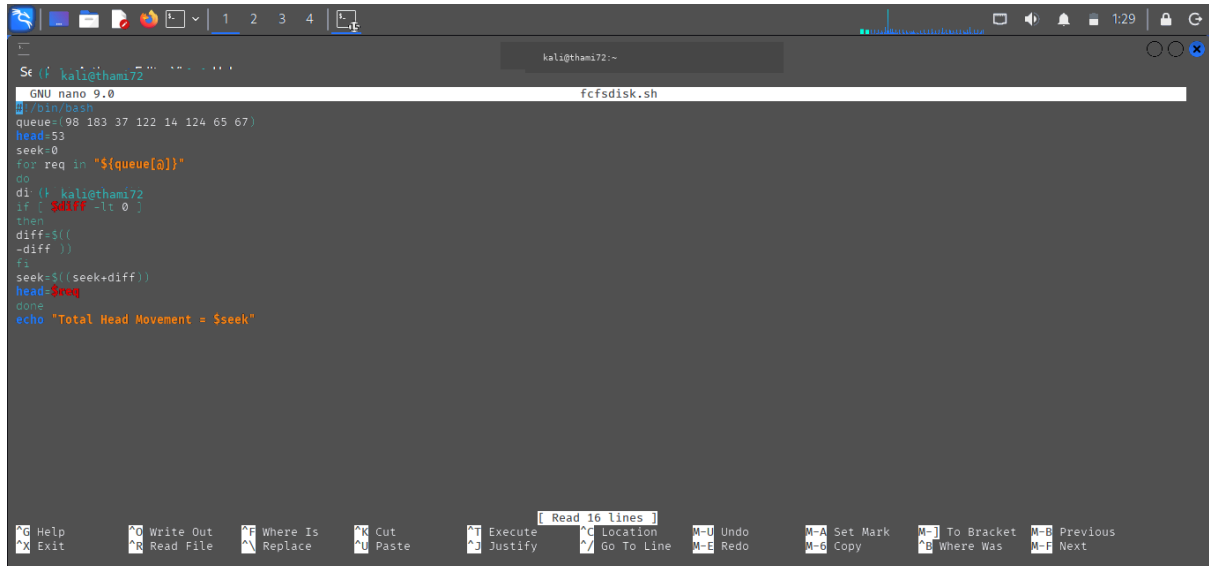
```
GNU nano 9.0 fcfsdisk.c
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int req[20], n, head, i;
    int seek = 0;
    printf("Enter Number of Requests: ");
    scanf("%d", &n);
    printf("Enter Request Queue:\n");
    for(i=0; i<n; i++)
        scanf("%d", &req[i]);
    printf("Enter Initial Head Position: ");
    scanf("%d", &head);
    for(i=0; i<n; i++)
    {
        seek += abs(req[i] - head);
        head = req[i];
    }
    printf("Total Head Movement = %d\n", seek);
    return 0;
}
```

OUTPUT :



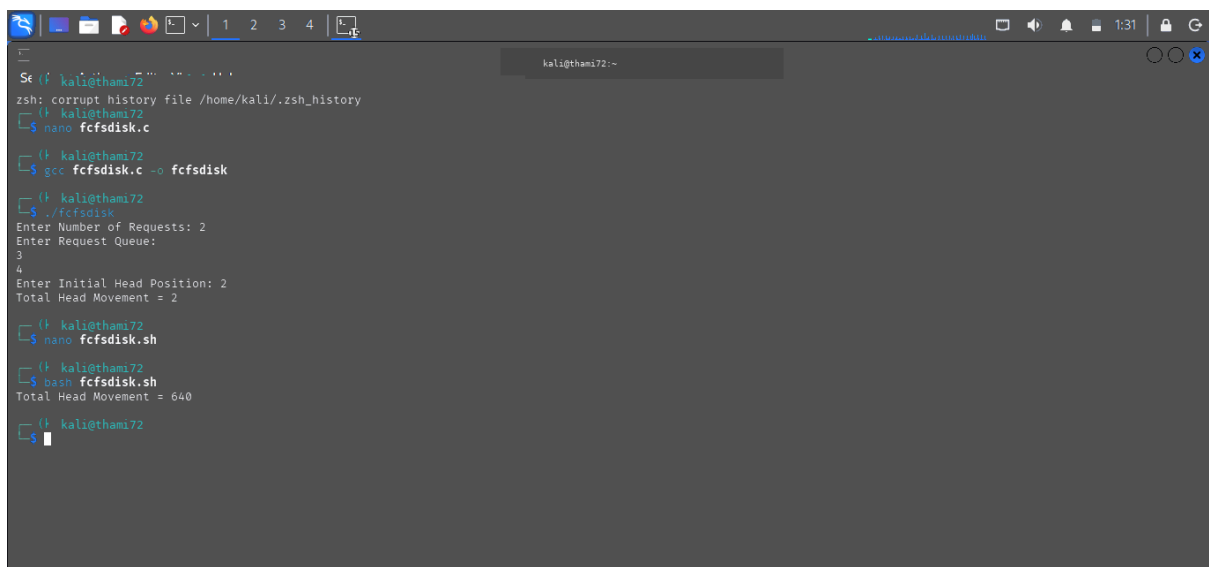
```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72:~$ nano fcfsdisk.c
kali@thami72:~$ gcc fcfsdisk.c -o fcfsdisk
kali@thami72:~$ ./fcfsdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 2
Total Head Movement = 2
kali@thami72:~$
```

SHELL PROGRAMMING : FCFS DISK



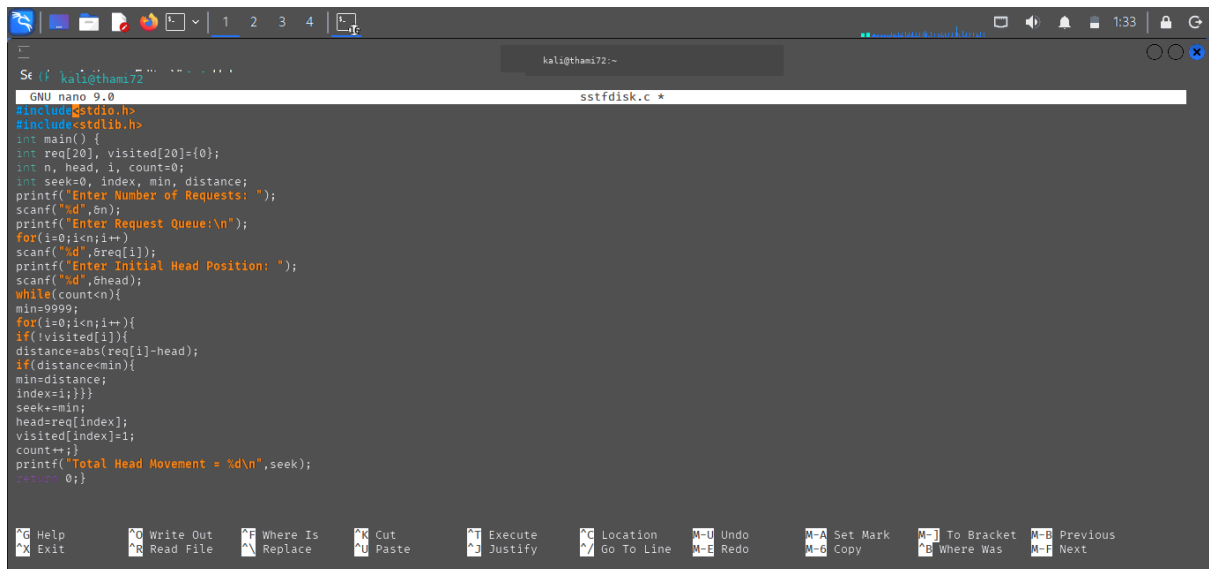
```
GNU nano 9.0 fcfdisk.sh
#!/bin/bash
queue=(98 183 37 122 14 124 65 67)
head=53
seek=0
for req in "${queue[@]}"
do
    di=$(kali@thami72)
    if [ $di -lt 0 ]
    then
        diff=$((
            -diff ))
    fi
    seek=$((seek+diff))
    head=$req
done
echo "Total Head Movement = $seek"
```

OUTPUT :



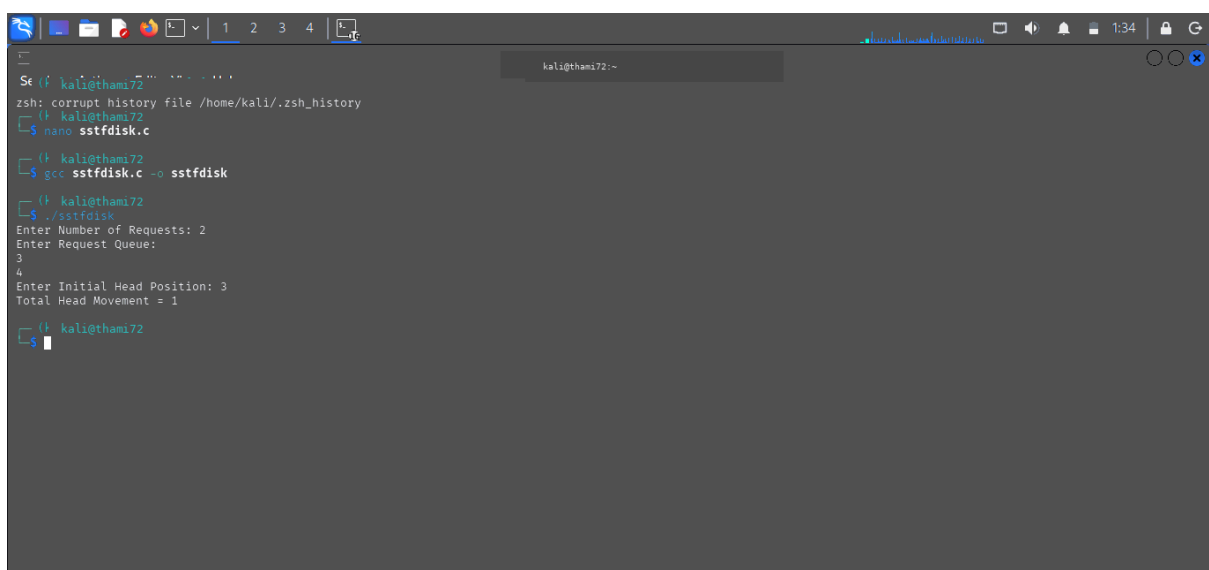
```
kali@thami72:~
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano fcfdisk.c
kali@thami72 ~
$ gcc fcfdisk.c -o fcfdisk
kali@thami72 ~
$ ./fcfsdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 2
Total Head Movement = 2
kali@thami72 ~
$ nano fcfdisk.sh
kali@thami72 ~
$ bash fcfdisk.sh
Total Head Movement = 640
kali@thami72 ~
$
```

C PROGRAMMING : SSTF DISK



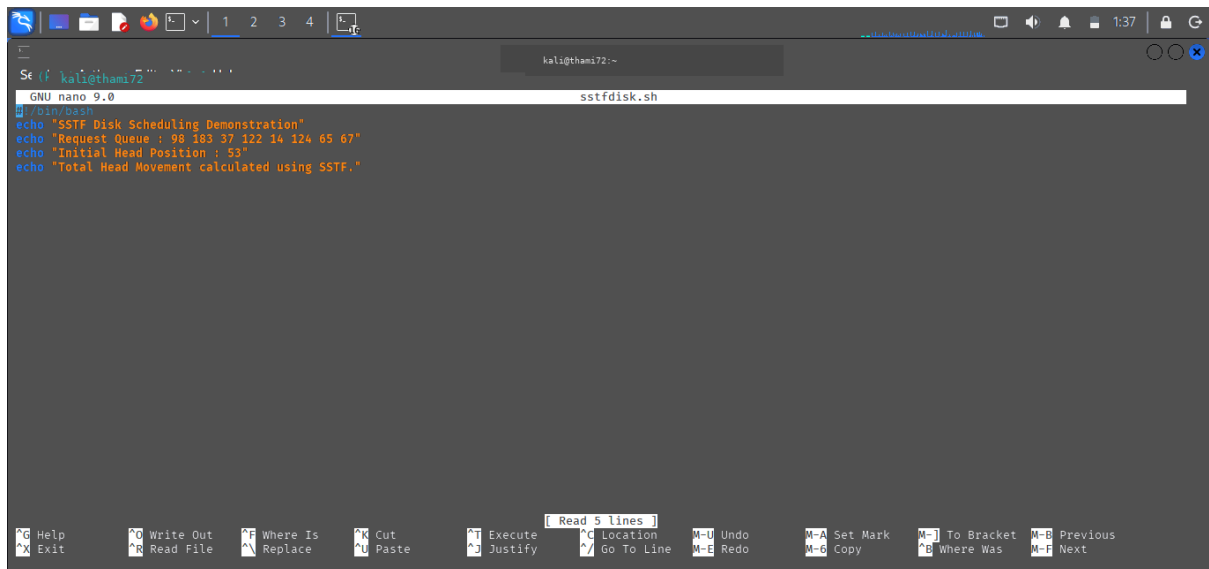
```
GNU nano 9.0 sstfdisk.c *
#include<stdio.h>
#include<stdlib.h>
int main() {
    int req[20], visited[20]={0};
    int n, head, i, count=0;
    int seek=0, index, min, distance;
    printf("Enter Number of Requests: ");
    scanf("%d",&n);
    printf("Enter Request Queue:\n");
    for(i=0;i<n;i++){
        scanf("%d",&req[i]);
        printf("Enter Initial Head Position: ");
        scanf("%d",&head);
        while(count<n){
            min=9999;
            for(i=0;i<n;i++){
                if(!visited[i]){
                    distance=abs(req[i]-head);
                    if(distance<min){
                        min=distance;
                        index=i;
                    }
                }
            }
            seek+=min;
            head=req[index];
            visited[index]=1;
            count++;
        }
        printf("Total Head Movement = %d\n",seek);
        return 0;
    }
```

OUTPUT :



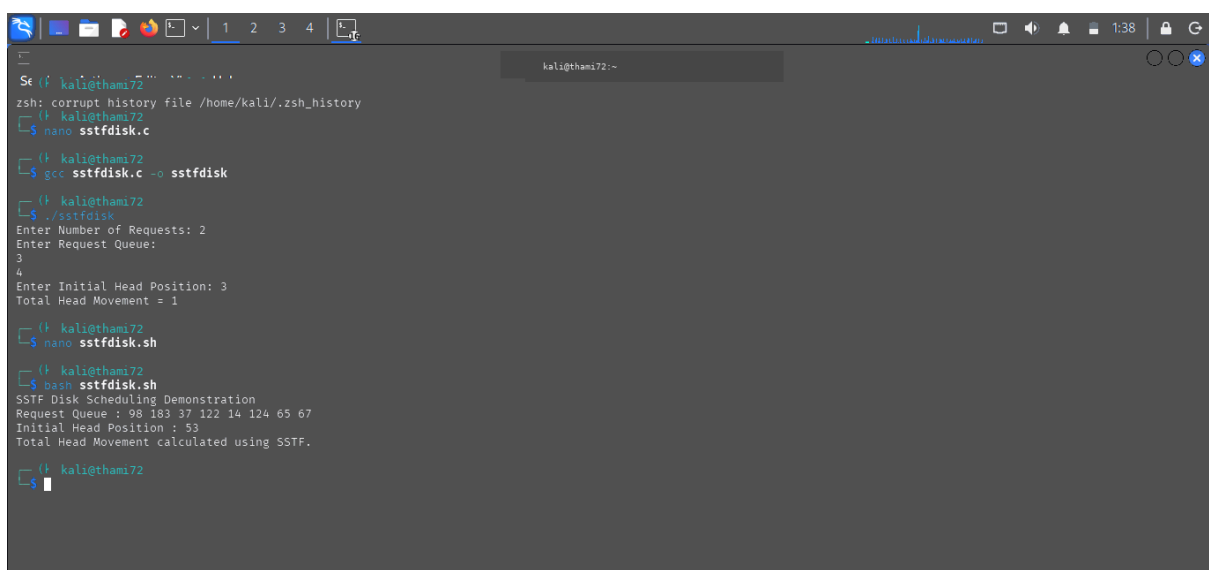
```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72:~$ nano sstfdisk.c
kali@thami72:~$ gcc sstfdisk.c -o sstfdisk
kali@thami72:~$ ./sstfdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 3
Total Head Movement = 1
kali@thami72:~$
```

SHELL PROGRAMMING : SSTF DISK



```
GNU nano 9.0 sstfdisk.sh
#!/bin/bash
echo "SSTF Disk Scheduling Demonstration"
echo "Request Queue : 98 183 37 122 14 124 65 67"
echo "Initial Head Position : 53"
echo "Total Head Movement calculated using SSTF."
```

OUTPUT :



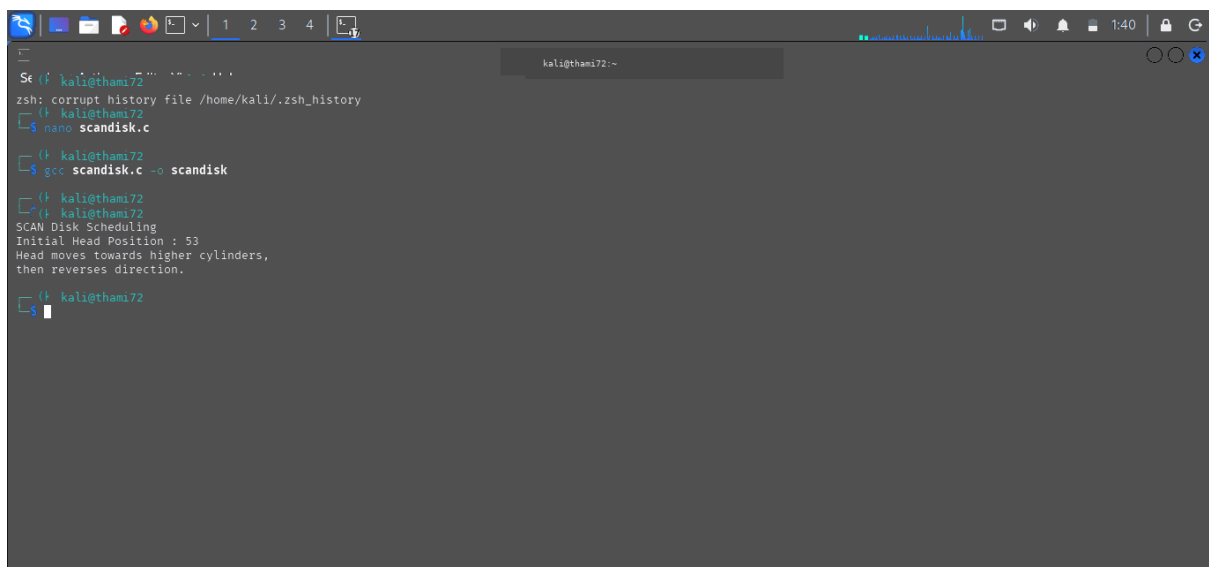
```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72:~$ nano sstfdisk.c
kali@thami72:~$ gcc sstfdisk.c -o sstfdisk
kali@thami72:~$ ./sstfdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 3
Total Head Movement = 1
kali@thami72:~$ nano sstfdisk.sh
kali@thami72:~$ bash sstfdisk.sh
SSTF Disk Scheduling Demonstration
Request Queue : 98 183 37 122 14 124 65 67
Initial Head Position : 53
Total Head Movement calculated using SSTF.
kali@thami72:~$
```

C PROGRAMMING : SCAN DISK



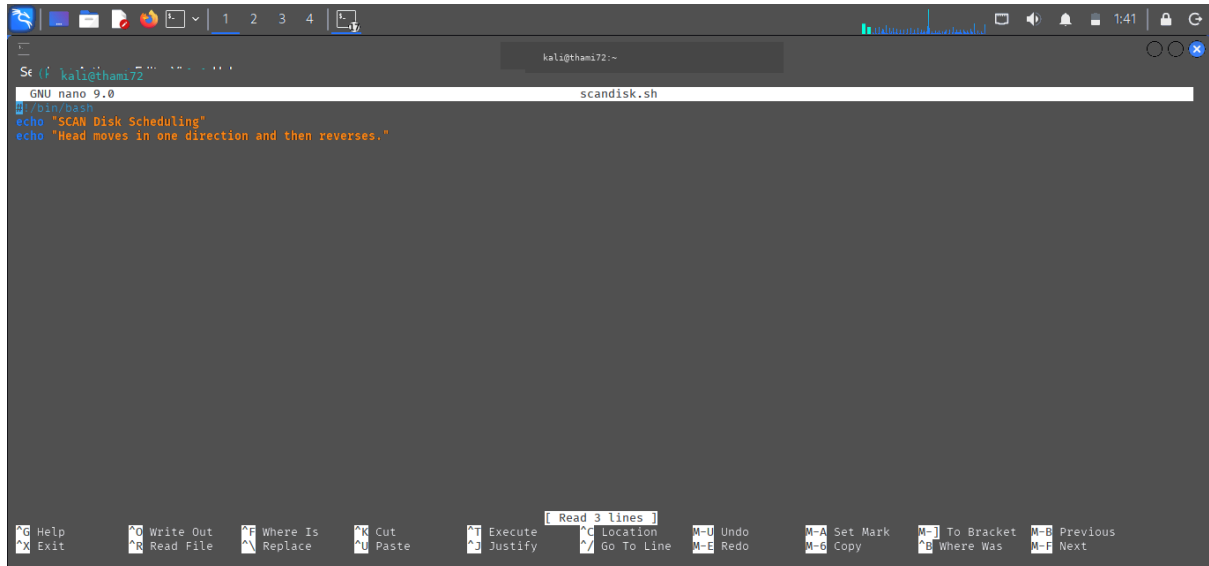
```
GNU nano 9.0 scandisk.c
#include<stdio.h>
int main()
{
    int disk_size = 200;
    int head = 53;
    printf("SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
    printf("Head moves towards higher cylinders,\n");
    printf("then reverses direction.\n");
    return 0;
}
```

OUTPUT :



```
Se (f kali@thami72 ~)
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano scandisk.c
kali@thami72 ~
$ gcc scandisk.c -o scandisk
kali@thami72 ~
$ ./scandisk
SCAN Disk Scheduling
Initial Head Position : 53
Head moves towards higher cylinders,
then reverses direction.
kali@thami72 ~
$
```

SHELL PROGRAMMING : SCAN DISK

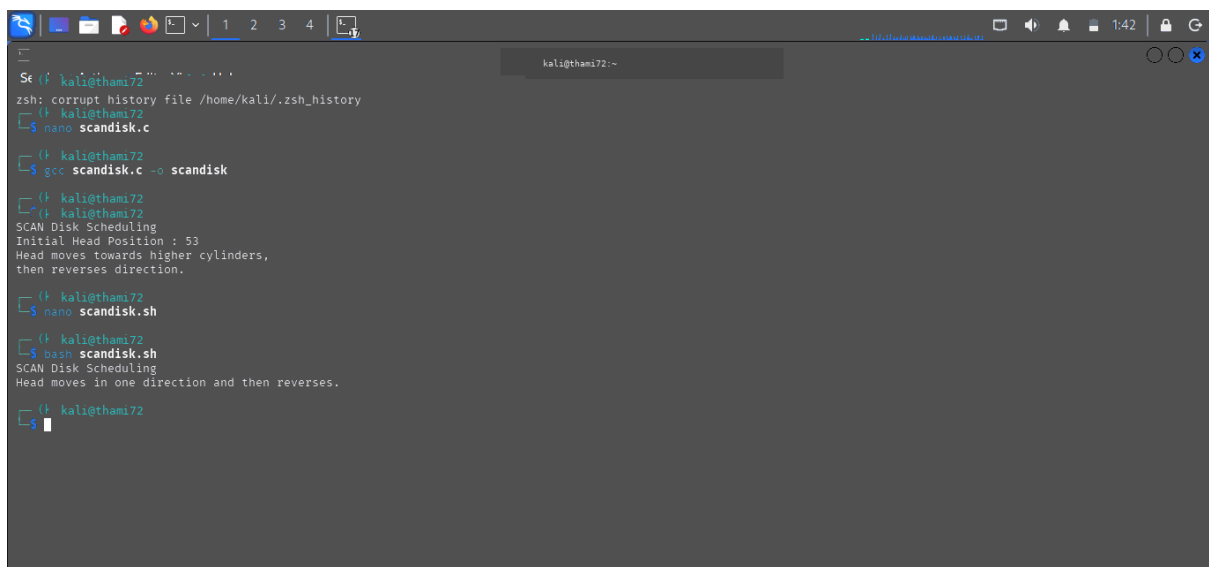


A terminal window titled 'kali@thami72:~' shows the nano 9.0 editor editing a file named 'scandisk.sh'. The editor's status bar at the top indicates 'GNU nano 9.0 scandisk.sh'. The file content is as follows:

```
#!/bin/bash
echo "SCAN Disk Scheduling"
echo "Head moves in one direction and then reverses."
```

The bottom of the terminal displays the nano editor's command shortcuts: Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Read 3 lines, Location, Go To Line, Undo, Redo, Set Mark, Copy, To Bracket, Where Was, Previous, Next.

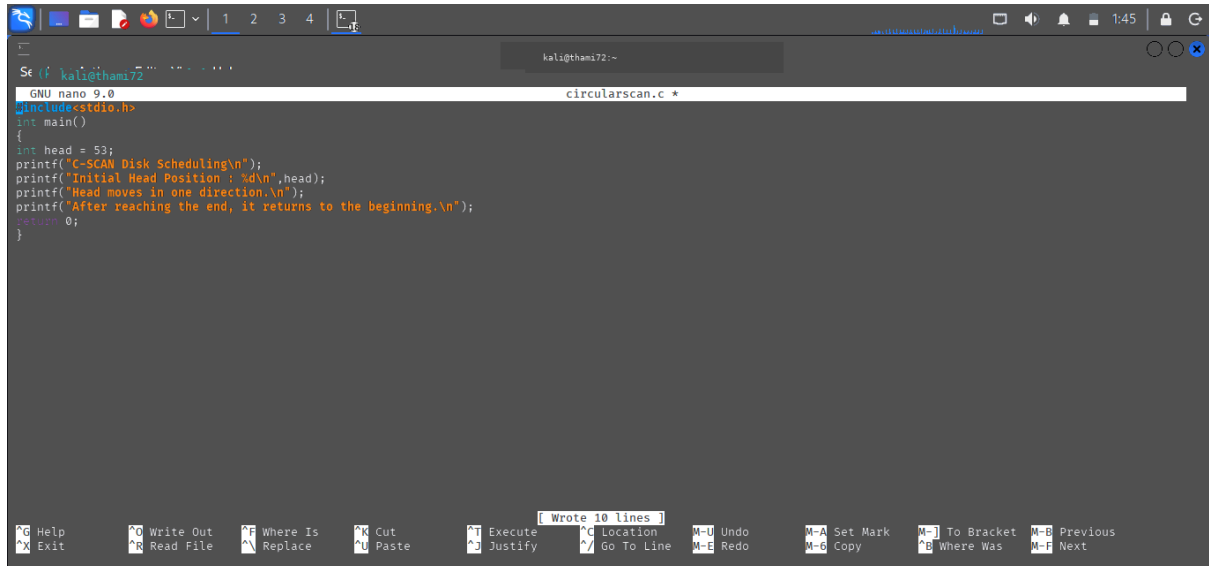
OUTPUT :



A terminal window titled 'kali@thami72:~' shows the execution of the scandisk.sh script. The terminal output is as follows:

```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano scandisk.c
kali@thami72 ~
$ gcc scandisk.c -o scandisk
kali@thami72 ~
$ ./scandisk
SCAN Disk Scheduling
Initial Head Position : 53
Head moves towards higher cylinders,
then reverses direction.
kali@thami72 ~
$ nano scandisk.sh
kali@thami72 ~
$ bash scandisk.sh
SCAN Disk Scheduling
Head moves in one direction and then reverses.
kali@thami72 ~
$
```

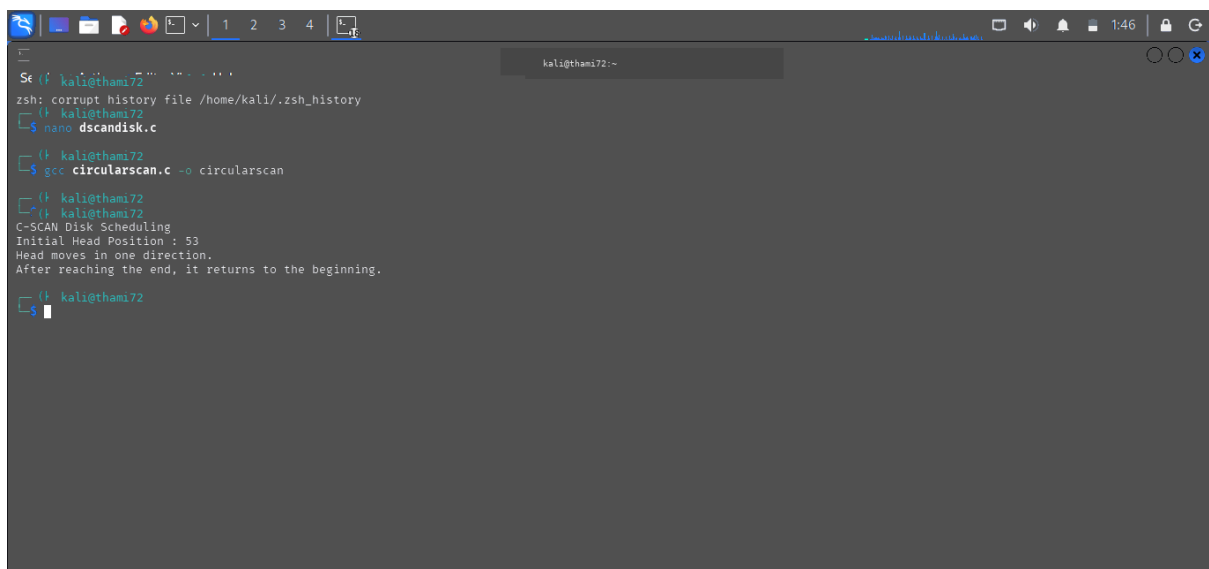
C PROGRAMMING : CIRCULAR SCAN



```
GNU nano 9.0 circularscan.c *
#include<stdio.h>
int main()
{
    int head = 53;
    printf("C-SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
    printf("Head moves in one direction.\n");
    printf("After reaching the end, it returns to the beginning.\n");
    return 0;
}
```

The screenshot shows a terminal window with the nano text editor open. The file being edited is circularscan.c. The code defines a main function that sets a head position of 53 and prints several messages about the C-SCAN disk scheduling algorithm. The terminal window title is 'kali@thami72:~'.

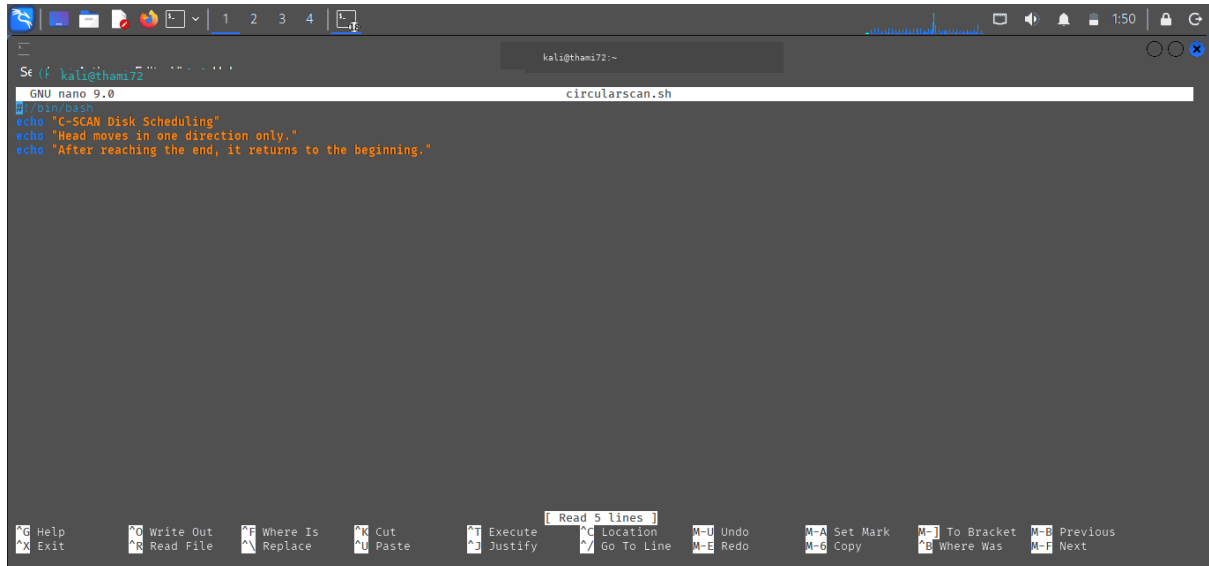
OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano dscandisk.c
kali@thami72 ~
$ gcc circularscan.c -o circularscan
kali@thami72 ~
$ ./circularscan
C-SCAN Disk Scheduling
Initial Head Position : 53
Head moves in one direction.
After reaching the end, it returns to the beginning.
kali@thami72 ~
$
```

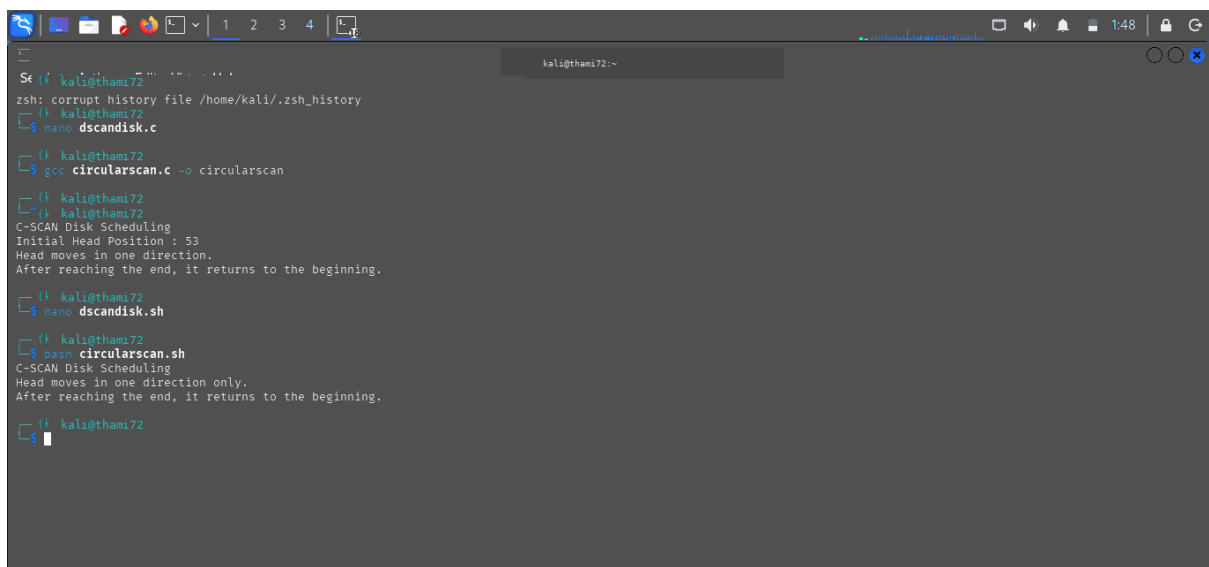
The screenshot shows the terminal output after running the program. It displays the compilation command 'gcc circularscan.c -o circularscan' and the execution command './circularscan'. The output of the program matches the printf statements in the code: 'C-SCAN Disk Scheduling', 'Initial Head Position : 53', 'Head moves in one direction.', and 'After reaching the end, it returns to the beginning.'.

SHELL PROGRAMMING : CIRCULAR SCAN



```
Se (f kali@thami72 ~)
GNU nano 9.0 circularscan.sh
#!/bin/bash
echo "C-SCAN Disk Scheduling"
echo "Head moves in one direction only."
echo "After reaching the end, it returns to the beginning."
```

OUTPUT :



```
Se (f kali@thami72 ~)
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano dscandisk.c
kali@thami72 ~
$ gcc circularscan.c -o circularscan
kali@thami72 ~
$ bash circularscan.sh
C-SCAN Disk Scheduling
Initial Head Position : 53
Head moves in one direction.
After reaching the end, it returns to the beginning.
kali@thami72 ~
$
```

